

Hotel Reputation Analytics — End-to-End BI Project

Solo project | SQL · Power BI · AI-assisted analytics | ~35,900 guest reviews, 626 hotels

I took a raw dataset of roughly 35,900 hotel guest reviews and turned it into a decision-ready Power BI dashboard and stakeholder report — owning the full pipeline from raw database to final deliverable.

What I did

Loaded & validated the data. Imported a ~35,900-row review file into a MySQL database. When the standard import silently loaded only 764 rows, I diagnosed the truncation, scripted a reliable bulk load instead, verified the full row count, and fixed character-encoding issues so international hotel names rendered correctly.

Caught a critical data-quality issue. While cleaning, I noticed the rating column didn't behave like a 1–5 scale — some scores reached 10. I investigated and found the source had **mixed a 1–5 and a 1–10 rating scale**, traced the off-scale ratings to a specific handful of hotels, and isolated them — leaving a clean 1–5 dataset of ~33,800 rated reviews. Without catching this, every average rating would have been wrong.

Used AI to mine 35,000 rows of free text. Rather than analysing star ratings alone, I used **AI-assisted text analysis** to read the written review content at scale, detect recurring complaint themes (cleanliness, bathrooms, odour, noise, pests), and quantify them into a chart — turning thousands of rows of unstructured text into a ranked view of *why* guests were unhappy.

Built the data model & dashboard. Created the measures and calculated tables (average rating, detractor rate, year-over-year change, rating bands, consistency), defined relationships, and designed a three-page Power BI dashboard: executive overview, reputation trends, and a brand / root-cause deep dive.

Delivered for stakeholders. Produced a polished PDF report and a one-page executive brief, framing findings as positives, negatives and averages with prioritised, actionable recommendations.

Key outcome

I showed that weak reputation was concentrated in budget brands and driven overwhelmingly by cleanliness and room-condition issues, and pinpointed the exact hotels to fix first — making the analysis directly actionable rather than just descriptive.

Tools & skills

SQL (MySQL)	Power BI	DAX	Data cleaning & validation
Text analysis	AI-assisted analytics	Data storytelling	

Hotel Reputation Analysis

What guests are telling us, where we stand, and what to act on

Stakeholder report | Based on ~33,800 guest reviews across 626 rated hotels, 2011–2016 | Reputation analysis (no financial data)

Executive summary

Overall guest satisfaction sits at **3.76 out of 5** and has been **flat and stable for years** — there is no broad decline to worry about. The issues are not spread evenly; they are **structural**. Weak reputation concentrates in specific **budget brands**, and the complaints behind it are overwhelmingly about **cleanliness and the physical condition of rooms**. At the other end, upscale brands and a handful of standout independent hotels perform consistently well. In short: this is a **targeted problem with a targeted fix**, not a portfolio-wide crisis.

33,783 REVIEWS ANALYSED	626 HOTELS EVALUATED	3.76 / 5 OVERALL GUEST RATING	14 PRIORITY PROBLEM HOTELS
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Top recommended actions

- **Run a cleanliness program across the budget chains** (Super 8, Days Inn, Quality Inn). One brand-level initiative reaches hundreds of properties at once — the single highest-leverage move available.
- **Triage the worst, busiest hotels first.** Fiesta Inn and Suites (1.46 stars over 533 reviews) is the clearest priority; nearly 9 in 10 of its guests leave a 1–2 star review.
- **Stop the bleeding** at hotels that are already weak *and* still falling — Crossland Economy Studios and Red Roof Inn Cedar Rapids appear on both lists.
- **Bottle what the winners do.** Document the playbook from the upscale brands and the fastest-recovering hotel (Super 8 Kansas City Airport, up almost a full star in a year) and spread it.

What we analysed, and how

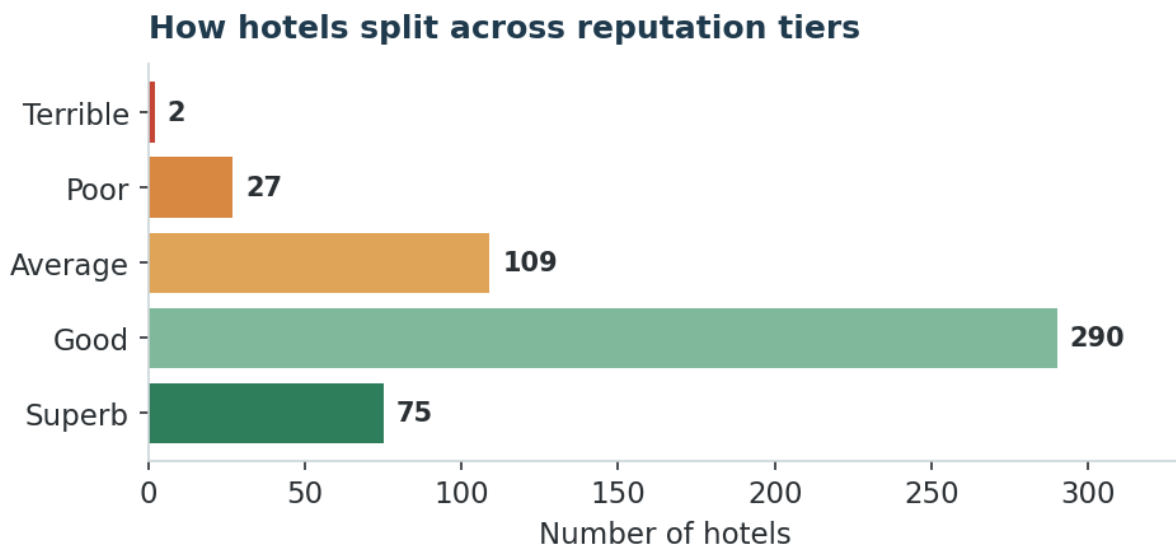
The analysis is built on a dataset of guest reviews. Before drawing any conclusions, the raw data was cleaned and structured so the findings would be trustworthy:

- **Removed duplicates** — 304 repeated reviews that would otherwise double-count.
- **Fixed a mixed rating scale** — the source mixed a 1–5 and a 1–10 scale. The affected hotels were isolated and set aside, and ~300 junk records (rated 0) were removed, so every rating now sits cleanly on a 1–5 scale.
- **Set a fairness threshold** — only hotels with 10+ reviews are ranked, so a single angry or delighted guest can't distort a hotel's standing.
- **Classified every hotel** into five reputation tiers — Superb, Good, Average, Poor, Terrible — using a Booking.com-style scale.

Scope note: the data contains no pricing or revenue figures, so this is a reputation analysis — it does not draw conclusions about profit. Location fields were found unreliable and were excluded.

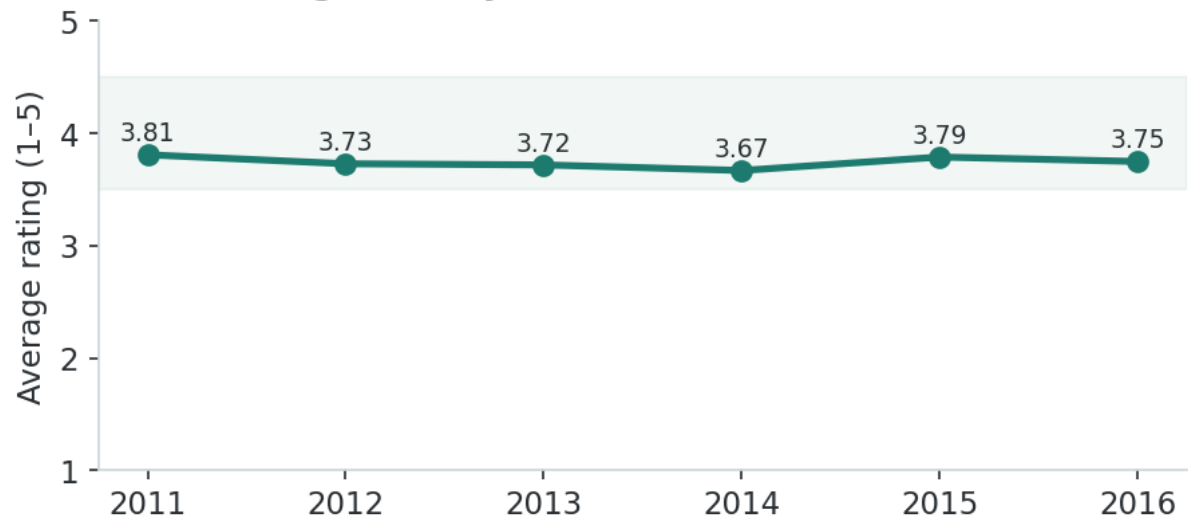
The overall picture

Most well-reviewed hotels are healthy: roughly three-quarters sit in the Good or Superb tiers. A meaningful minority — 29 hotels in the Poor or Terrible tiers — is where attention is needed.



Over time, reputation is steady. The chart below uses the full 1–5 scale so the trend is shown honestly: the year-to-year movement is a fraction of a star. Note too that about **77% of all reviews come from 2015–2016**, so the recent picture is the most reliable.

Guest rating has stayed flat and stable



Positives — what's working

The standout hotels (50+ reviews)

These are reliably excellent — high ratings backed by real volume. They are the internal models to learn from.

Hotel	Avg rating	Reviews
Villa Carlotta	4.90	50
Excellence Riviera Cancun	4.88	64
Simpson House Inn	4.78	125
Cliffrose Lodge & Gardens	4.73	157
The Alexandrian, Autograph Collection	4.48	1,185

The winning brands

Upscale and premium-select brands lead the field — Residence Inn, Doubletree, Hampton Inn, Country Inn and Holiday Inn all average around 4 stars or better. The brand chart on the next page shows the full picture.

Hotels on the way up (2015 → 2016)

Hotel	Rating change
Super 8 Kansas City Airport	+0.98
Super 8 Bradenton / Sarasota	+0.74
Baymont Inn & Suites Boone	+0.73
Springhill Suites Columbia	+0.62

That a Super 8 can gain almost a full star in a year proves the budget-tier problem is fixable, not inevitable.

Negatives — what needs attention

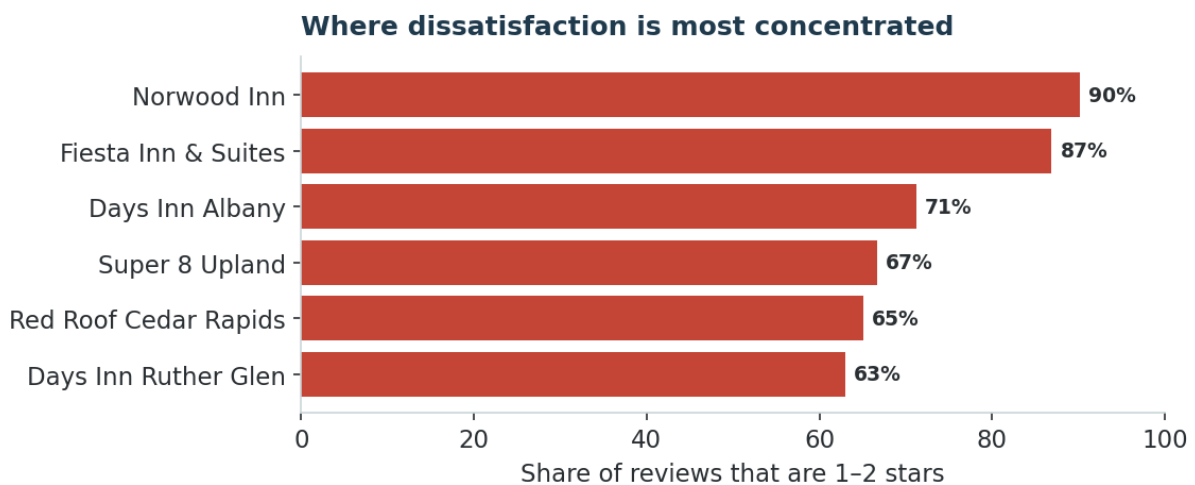
The priority fixes (50+ reviews)

Poor or Terrible hotels with enough reviews to be certain — ranked by how many guests are affected.

Hotel	Avg rating	Reviews
Fiesta Inn and Suites	1.46	533
Days Inn and Suites Albany	2.03	111
Red Roof Inn Cedar Rapids	2.19	106
Norwood Inn and Suites	1.39	61
Super 8 – Upland	2.13	63

Where dissatisfaction is most concentrated

Detractor rate — the share of a hotel's reviews that are 1–2 stars — exposes the worst guest experiences even more sharply than the average. At Norwood Inn, **90% of all reviews are 1–2 stars**.



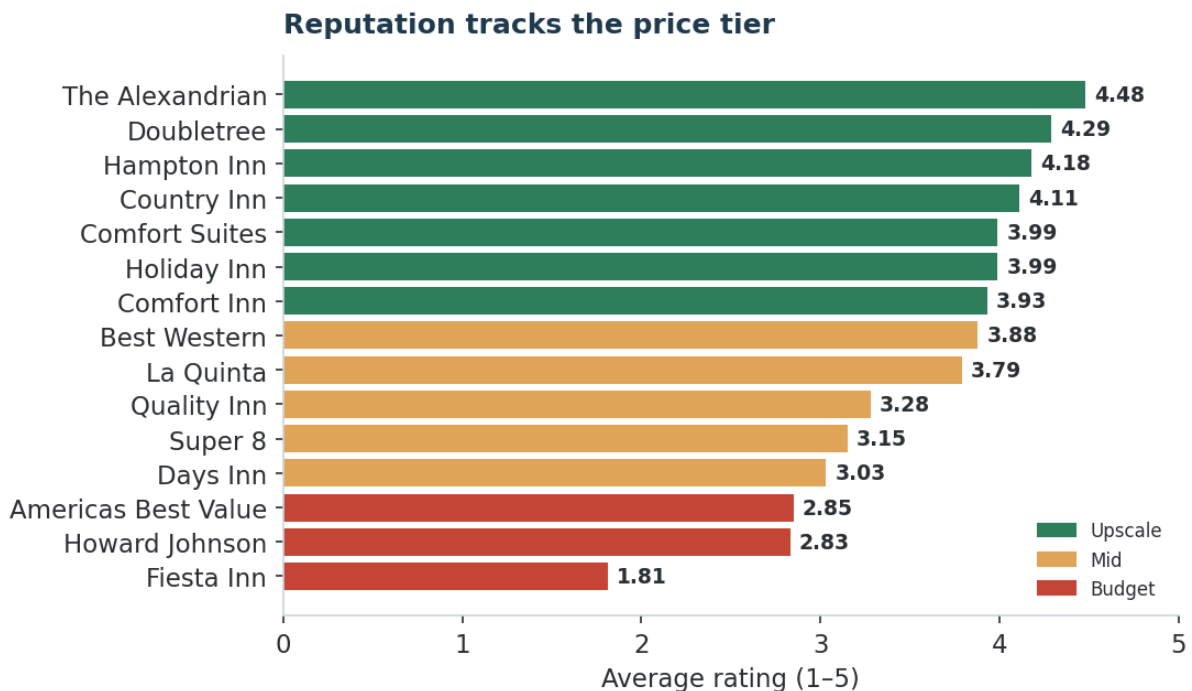
Hotels losing ground (2015 → 2016)

Hotel	Rating change
Crossland Economy Studios – Lake Charles	-0.99
Holiday Inn Express Hesperia	-0.91
Super 8 Metropolis	-0.82
Days Inn & Suites Castle Rock	-0.64

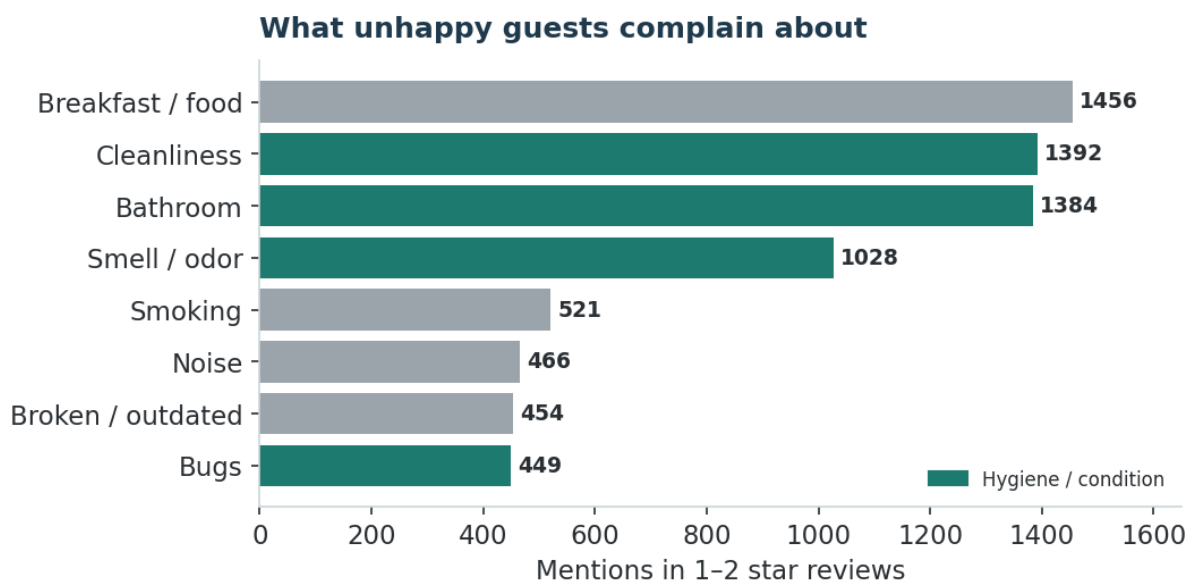
Crossland Lake Charles and Red Roof Cedar Rapids are the most urgent of all — already weak and still declining.

Why — the root cause

Reputation tracks the price tier almost perfectly. Budget brands underperform *as a group*, which is exactly why a brand-level response works better than chasing individual hotels.



Reading the actual complaints in the 1–2 star reviews shows what drives the anger. Food and breakfast come up most often, but the dominant story is **hygiene and physical condition**: cleanliness, bathrooms, smell and bugs together far outweigh everything else. Guests forgive 'basic'; they do not forgive 'dirty'.



Average — the stable baseline

Around 109 well-reviewed hotels sit in the Average tier. They meet expectations without standing out, and need no intervention — effort is far better spent lifting the Negatives and replicating the Positives. They are worth keeping an eye on only as a group, to catch any drift early.

What to do next

- **Cleanliness first, brand-wide.** A standardised cleanliness and room-condition standard across the budget chains addresses the single biggest driver of bad reviews, at scale.
- **Triage the worst high-volume hotels** individually — starting with Fiesta Inn and Suites — with on-site inspections.
- **Put the decliners on a watch-list** and intervene before they hit the bottom tier.
- **Capture and spread the winners' playbook**, including the fast-improving hotels, so good practice doesn't stay siloed.
- **Track detractor rate going forward**, not just the average — it catches a struggling hotel sooner.

Reputation analysis prepared from guest-review data. Figures reflect hotels with valid, single-scale ratings; rankings use a 10+ review threshold. No financial data was available, so no profit or revenue conclusions are drawn.